TREND//RADAR – YouTube Trend Intelligence | Components Map

From live YouTube data to three outputs (Live Radar, Trend Intelligence, Sunday Brief email)

Actor / Tool	Trigger	Input	Context	AI	Processing	Output
Creator (Ishaan)	**Ishaan runs the radar** Clicks Run Radar on the site, any time he wants a fresh read	**His channel settings** Categories (AI tools, productivity, tech reviews), region India, 7-day look-back				**Films on Monday** Knows what to make, why now, the format, angle, title and length
Vercel Cron (Scheduler)	**Sunday 08:00 IST** Weekly scan + brief; a daily 06:00 job refreshes view counts					
YouTube (Data API v3)		**Live YouTube videos** Keyword search per category + India Science & Tech trending chart; views, likes, comments, duration (≈600 quota units)				
Supabase (Database)			**Saved context** • Settings • Daily view snapshots (history) • Past scans and briefs			**History saved** Run, topics, videos, snapshots and the brief are stored for next week
Gemini Flash (AI)				**Gemini Flash finds the topics** Groups videos into 3-5 topic clusters, labels formats, writes why-now, angle, title and the brief. Zod-checked, retried once		
TREND//RADAR (Engine)			**Engine prepares the data** Engagement, views/hour, upload momentum; drops Shorts; sends Gemini a compact 90-video table		**Engine scores each topic** 0-100 trend score from 6 weighted factors, Rising / Stable / Cooling, confidence; saves the run	
Vercel (Website)						**Live Radar + Trends + Brief** Scroll story with rotating topics, 3-5 topics with evidence, score and what to make
Email (Resend)						**Sunday Brief email** Arrives before noon IST with the Monday filming plan

Solid arrow = Ishaan's own step Dashed arrows = data passed between tools **Every number is calculated by the app.** Gemini Flash only writes words. No history yet → shows **BUILDING BASELINE**, never invented data